Copilot in Excel: FMEA and risk prioritisation

100 minutes

IES | Original synthetic CW-401 engineering case. Values and limits are fictional teaching assumptions, not plant operating instructions. Generated references are independently calculated examples, not recorded Copilot executions.

Preparation

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged.

1. Open the shared case

Use your previous workbook or this lab's standalone Starter.xlsx. Read Evidence, FMEA_Input and ScoringNote. Reuse the SPC interpretation if available; no completed Lab01 file is required.

2. Create the review table

Select FMEA_Input; submit FMEA1. Check all six IDs and columns. FM01 has observed low-delivery evidence but an unconfirmed restriction cause; FM04 is a prospective leakage scenario with no supplied leak event. Keep source status explicit.

3. Validate the ratings

Read the supplied provisional scale definitions. Higher D means poorer detectability. Every rating is a workshop judgement and must retain its rationale. Do not recalculate occurrence from 120 tests or treat a missing incident as proof of low risk.

4. Use formulas and charts

Submit FMEA2. RPNs: FM01=140; FM02=70; FM03=108; FM04=90; FM05=48; FM06=32. Sorted order is01,03,04,02,05,06. Highlight FM04 separately because S=9. Verify formulas remain attached to IDs after sorting and create an RPN column chart.

5. Run a short evidence-led debate

Submit FMEA3. Reliability examines mechanism evidence, operations examines practicality, safety examines consequence and approval gaps. Each challenges one claim; moderator retains unresolved gaps. These are simulated roles, not independent experts. The removal of a strainer is not effectiveness evidence for every failure mode.

6. Create an action view

Keep proposed actions separate from current controls. Add owner role, due-date proposal, required effectiveness evidence and status. Leave residual S/O/D blank or explicitly unassessed. Do not invent an implementation date or approved work instruction.

7. Save and peer review

Save FMEA_Review, priority chart and disagreement/evidence log. Exchange one row with a peer: identify function/mode/effect/cause/control and recompute its RPN. Explain why severity review can outrank the numerical ranking.

Copilot prompts

FMEA1 Structure the review

Use FMEA_Input and Evidence. Create FMEA_Review with function, mode, effect, potential cause, current control, evidence, supplied provisional S/O/D, RPN, action, owner and status. Keep the six mode IDs. Separate actual observations from prospective scenarios. Use ScoringNote for definitions and do not infer occurrence ratings from the short test series.

FMEA2 Calculate and prioritise

Add an Excel formula RPN=S*O*D for every row. Sort a view by RPN descending while preserving mode IDs and source rows. Add a separate high-severity review flag for S>=9. Create an RPN column chart. Do not treat rank or low RPN as evidence of safety acceptance.

FMEA3 Challenge with personas

Simulate reliability, operations and safety review lenses. First return one separate evidence-based concern per role. Run two concise challenge/response rounds, citing mode IDs and evidence records. A moderator returns agreed revisions, unresolved disagreement and missing evidence. Do not force consensus or invent approval. Keep residual ratings unassessed until action-effectiveness evidence exists.